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eiro (Brazil) to San Jorge  
*urys rasilis* is a small- to medium-sized flounder (20 cm) that feeds exclusively on benthic invertebrates, with small crabs constituting its primary prey (Díaz de Astarloa and Munroe 1998; Cousseau and Perrotta 2013). This species is of commercial importance, yet it represents about 6% of the flounder landings in the region (Fabr  and D az de Astarloa 2001; D az de Astarloa 2002; Rico 2010). *Xystreuryx rasilis* is most abundant in high-salinity waters at 50-100 m in the Buenos Aires Coastal Ecosystem (BCE) (D az de Astarloa 2002; D az de Astarloa and Fabr  2003) and at 50-90 m in San Mat as Gulf (SMG) (Perier and Di Gi acomo 2002) (Figure 1). According to the IUCN, *X. rasilis* is currently classified as a species of Least Concern (Riestra and D az de Astarloa 2020).

Within the BCE, two groups have been identified: one north of 36  S and another south of 37  S, the latter being considerably more abundant (Fabr 

et al. 2001). In addition, parasitological evidence indicates that at least two stocks of *X. rasilis* inhabit the coastal and shelf waters of the BCE: a northern group between 34  S and 38  S, and a southern group between 38  S and 41  S, in the area known as ‘El Rinc n’ (39  S-41  S) (Alarcos and Timi 2013). These stocks have been proposed as discrete management units to promote the sustainable exploitation of the resource (Alarcos and Timi 2013). Seasonal migrations have also been reported: in autumn, individuals move to deeper waters for feeding and growth, while in spring, mature specimens migrate toward the coast and further south (D az de Astarloa 2002). Both groups converge south of Mar del Plata (38  S) to reproduce (Fabr  et al. 2001). In these areas, variations in water masses generate temperature and salinity fronts with key biological functions (Acha et al. 2004), acting as breeding and nursery grounds for several fish species by enhancing larval retention (Piola and Rivas 1997). In addition, Alarcos and

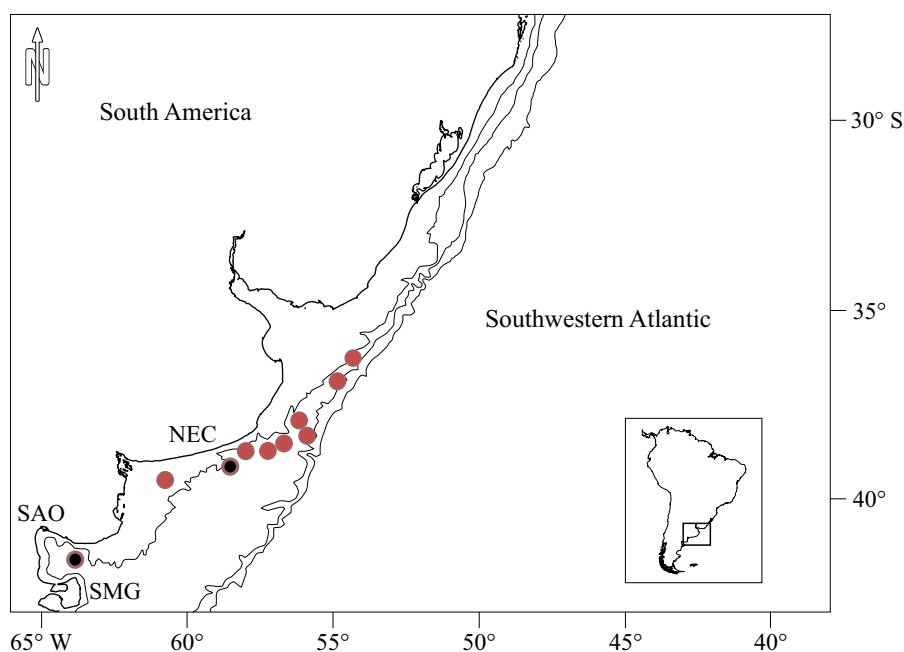


Figure 1. Main concentrations of *Xystreuryx rasilis* in the Buenos Aires Coastal Ecosystem (modified from Fabr  and D az de Astarloa 1996; D az de Astarloa and Fabr  2003), showing sampling sites (black circles). Lines on the continental shelf indicate depths of 50, 100, and 200 m. Necochea (NEC) and San Antonio Oeste (SAO) in San Mat as Gulf (SMG).

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ted incorporating samples from 'El Rincón' to assess whether the southern stock moves toward these latitudes during the warm season. The spawning area for this stock could be located at SMG, a north Patagonian gulf located in the southern limit of 'El Rincón' (Figure 1). This hypothesis is supported by the observed increase in the relative abundance and peak reproductive activity of *X. rasilis* in this gulf during the warm season (Perier and Di Giacomio 2002).

The aim of this study was to test the hypothesis that samples from Necochea ('El Rincón') and San Antonio Oeste (SMG) constitute distinct genetic units. Samples of *X. rasilis* were obtained from landings of the Argentine coastal fleet in the coastal waters off Necochea (NEC; 38° 40' S, 58° 37' W; 12 specimens) and from San Antonio Oeste (SAO) in SMG (41° 30' S, 64° 15' W; 7 specimens) (Figure 1). Muscle tissue samples were preserved in ethanol. Amplification of a mitochondrial control region fragment was performed using the universal primers developed by Lee et al. (1995): A (5' TTC-CACCTCTAACTCCCAAAGCTAG 3') and E (5' CCTGAAGTAGGAACCAGATG 3'), following the protocols described by Fainburg et al. (2022). Sequences were manually edited using Proseq v.2.91 (Filatov 2002) and aligned using ClustalX2 (Larkin et al. 2007). *Xystreuryx rasilis* sequences of the control region were deposited in GenBank (access no: PP506082- PP506065).

Haplotype diversity ( $h$ ) and nucleotide diversity ( $\pi$ ) for the mitochondrial control region fragment were estimated using DnaSP (Librado and

represent the distribution of pairwise differences among haplotypes (Rogers and Harpending 1992). This approach can discriminate between sudden population expansion and long-term stability. Expansion times and demographic changes in effective population size, weighted by generation time ( $Ne t$ ), were inferred using the Bayesian Skyline Plot model in BEAST 1.7 (Drummond et al. 2012). Analyses were performed under a relaxed molecular clock with an intra-specific substitution rate of 6%/Ma, as reported for marine fishes (see Fainburg et al. 2022, 2025). To account for uncertainty, results are also presented assuming mutation rates of 4%/Ma and 8%/Ma. Three independent Markov Chain Monte Carlo (MCMC) runs of 10 million steps each were performed, with trees and parameters sampled every 1,000 steps. Convergence was verified in Tracer, ensuring Effective Sample Sizes > 200. Only one representative run is shown, with 95% confidence intervals also reported (Drummond et al. 2012).

A 406-bp fragment of the mitochondrial control region was sequenced in *Xystreuryx rasilis* from Necochea (N = 12) and San Antonio Oeste (N = 7). Haplotype diversity was extremely high ( $h = 0.994$ ), and nucleotide diversity was also elevated at both sampling sites ( $\pi = 0.022$ ) (Table 1). Tajima's D was negative but not significant, whereas Fu's F test yielded a negative and highly significant value (Table 1), suggesting population expansion. Genetic divergence between NEC and SAO was low and not statistically significant ( $F_{ST} = 0.050$ ,  $p = 0.105$ ). Among the 19 individuals of *X. rasilis* analyzed, 18 distinct haplotypes were identified: 11 in NEC and 7 in SAO with one haplotype shared by two individuals from NEC (Figure 2). The haplotype network differed markedly from those reported for other flatfish species of the region (Fainburg et al. 2025), as no haplotypes were shared among populations, all haplotypes were separated by multiple mutational steps, and no clear phylogeographic structure was detected (Figure 2). *Xystreuryx rasilis* exhibited a demographic signal consistent with a bimodal pairwise mismatch distribution (Figure

were assessed with Tajima's D (Tajima 1983) and Fu's F (Fu 1997) using DnaSP (Ramos-Onsins and Rozas 2002). Demographic history was further investigated through mismatch distributions, which

nd demographic parameters of the Control Region in *Xystreureys rasilis* (406 bp). n = number of ber of haplotypes;  $h$  = haplotypic diversity;  $\pi$  = nucleotidic diversity; sd = standard deviation;  $D$  = Tajima's  $D$ ;  $F$  = Fu's  $F$ . \* $P < 0.05$ ; \*\* $P < 0.01$ . NEC: Necochea; SAO: San Antonio Oeste

| Site  | n  | Nh | $h \pm sd$        | $\pi \pm sd$      | $D$    | $F$      |
|-------|----|----|-------------------|-------------------|--------|----------|
| NEC   | 12 | 11 | $0.985 \pm 0.040$ | $0.023 \pm 0.002$ | -0.847 | -3.053*  |
| SAO   | 7  | 7  | $1.000 \pm 0.076$ | $0.019 \pm 0.003$ | -0.765 | -2.042*  |
| Total | 19 | 18 | $0.994 \pm 0.019$ | $0.022 \pm 0.002$ | -1.262 | -9.157** |

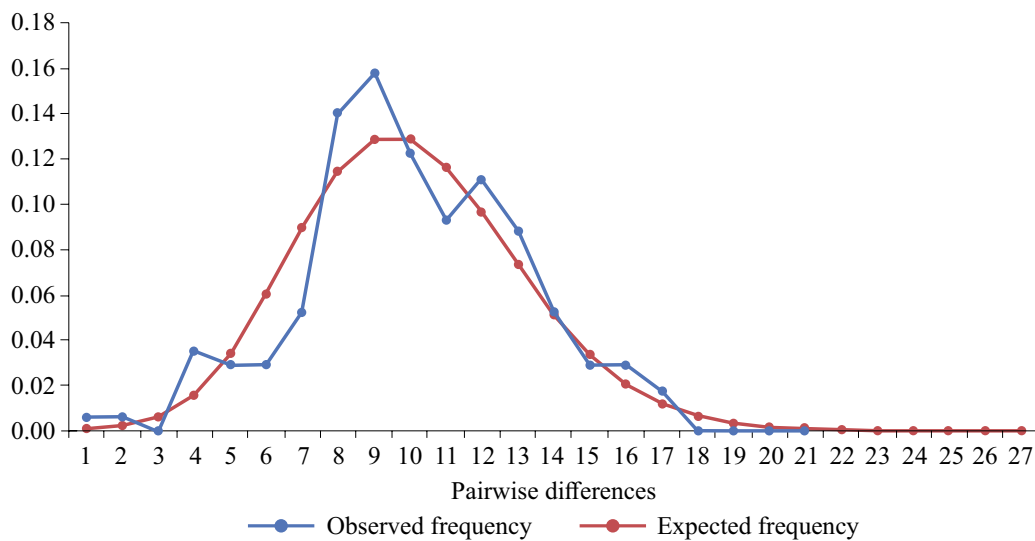


Figure 2. Mismatch distribution based on the mitochondrial control region of *Xystreureys rasilis*.

3), a pattern typically explained as evidence for two distinct lineages. In addition, fishes showing high haplotype and nucleotide diversities suggest a large and stable population with an extended evolutionary history, possibly resulting from historical isolation followed by secondary contact between differentiated allopatric lineages (Grant and Bowen 1998). Coalescent analyses further suggested a deep evolutionary history, consistent with long-term population stability with the most recent common ancestor dating to  $\sim 148,000$  years before present assuming a mutation rate of 6%/Ma (98,000-222,000 years before present considering mutation rates of 4%/Ma and 8%/Ma respective-

ly), accompanied by a gradual increase in effective population size over this period (Figure 4).

*Xystreureys rasilis* appears to have been little affected by the last glacial maximum (LGM) around 18,000-23,000 years ago, exhibiting high genetic variability, large effective population sizes, and an evolutionary history indicative of habitat stability. However, the haplotype network showed that most

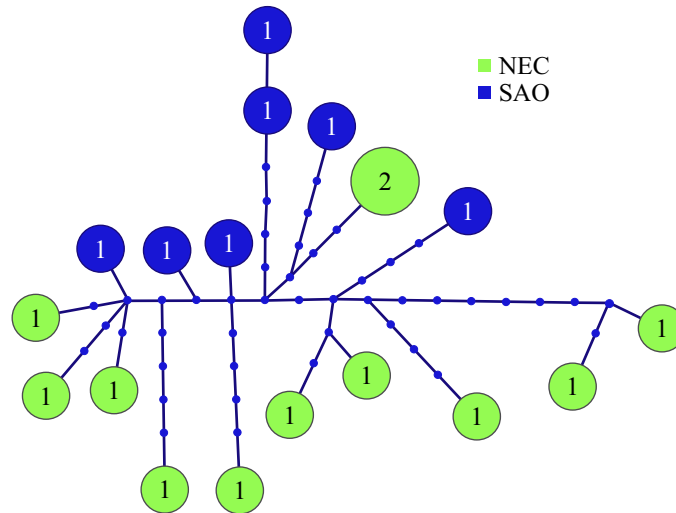


Figure 3. Mitochondrial DNA haplotype networks based on the control region of *Xystreuryx rasilis* from the Buenos Aires Coastal Ecosystem, including samples from Necochea (NEC) and San Antonio Oeste (SAO). Circle size represents haplotype frequency, and each line denotes a mutational step (dots) separating haplotypes.

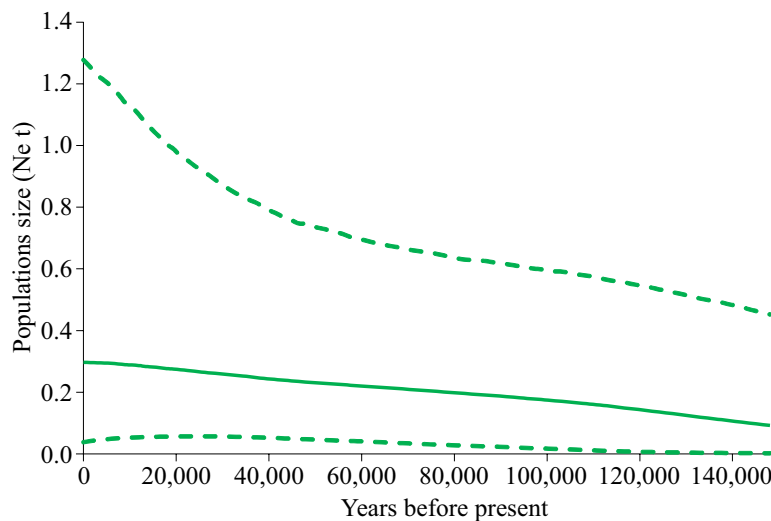


Figure 4. Bayesian Skyline Plot based on the mitochondrial control region of *Xystreuryx rasilis* showing the demographic history of the species. The y-axis represents the product of effective population size ( $N_e$ ) and generation time ( $t$ ). Dashed lines indicate 95% confidence intervals.

of *X. rasilis* and the post-LGM coastline habitats evolution. This pattern suggests that individuals from ‘El Rincón’ and the San Matías Gulf may have been historically isolated yet exhibit significant contemporary connectivity.

This genetic evidence should be interpreted

cautiously given the limited sample size; nevertheless, we must assume that these results, using only a mitochondrial marker, need to be further analyzed with a codominant marker such as micro-satellite loci or genomic SNP’s (such as RAD-seq technology).

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## EDGEMENTS

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### Data availability

DNA sequences generated in this study were submitted to GenBank. The control region sequences of *X. rasilis* were deposited in GenBank (access no: PP506082- PP506065).

### Conflict of interest

The authors declare no competing interests.

### Ethical approval

The authors declare that no ethical approval was required for the writing of this article.

### Author contributions

Leandro A. Fainburg: conceived and designed the experiment; generated data; conducted the pri-

mary data analysis; and contributed to the writing of the manuscript. David E. Sabadin: helped in data collection and wrote the initial draft of the manuscript. Pedro Fernández Iriarte: agreed to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved; drafted the final version of the manuscript. All authors reviewed and approved the manuscript.

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